

Complete Volume — All Three Unit Reports

Twelve Sections per Unit · Full Compilation (36 Sections)

GROUP 4 • Complete Volume — All Three Units • August 28, 2026

Educational mathematics assignment — not medical advice.

Master Table of Contents

This volume compiles the three standalone Unit reports back-to-back, each following the faculty's 12-section R&D format. The integrated full R&D report (GROUP4_RnD_Report) remains the concise 12-section synthesis; this volume is the expanded 36-section compilation for detailed submission.

- Part I — Unit I: Logic and Proofs (Sections 1–12, focused on propositional logic)
- Part II — Unit II: Combinatorics (Sections 1–12, focused on counting)
- Part III — Unit III: Graphs & Isomorphism (Sections 1–12, focused on bipartite graphs)

PART I

Unit I — Logic and Proofs

Twelve-section standalone report for Unit I, covering the Hypothetical Syllogism triage rule.

1. Title of the Case Study (Unit I)

Unit I — Logic and Proofs: Proving $(A \wedge B) \rightarrow T$ from $(A \wedge B) \rightarrow X$ and $X \rightarrow T$ for Symptom-Based Triage.

2. Problem Statement (Unit I)

Given $P_1: (A \wedge B) \rightarrow X$ and $P_2: X \rightarrow T$, does $(A \wedge B)$ entail T ? Must be proved via Hypothetical Syllogism and verified by truth table.

- If $A \wedge B$ true, can we derive X then T via Modus Ponens?
- If $A \wedge B$ false, does the implication hold vacuously?

3. Real-world Context

Triage chatbots need auditable rules: “two red flags together trigger a workup” must follow from the knowledge base, not be an ad-hoc shortcut.

4. Mathematical Concept Used

Propositions A, B, X, T ; connectives $\wedge, \rightarrow$; Modus Ponens, Conditional Proof, Hypothetical Syllogism; 16-row truth-table semantics.

5. Mathematical Formulation

Variables: A="symptom A present", B="symptom B present", X="disease X suspected", T="further testing recommended". Premises: $P_1: (A \wedge B) \rightarrow X$, $P_2: X \rightarrow T$. Theorem: $\vdash (A \wedge B) \rightarrow T$.

6. Analysis / Solution — Proof

Step	Statement	Justification
1	$(A \wedge B) \rightarrow X$	Premise P_1
2	$X \rightarrow T$	Premise P_2
3	$A \wedge B$	Assumption
4	X	MP 1,3
5	T	MP 2,4
6	$(A \wedge B) \rightarrow T$	CP 3-5 ■

7. Diagram / Truth Table

16-row table. Shaded rows: both premises true (1 fired + 9 vacuous).

A	B	X	T	$A \wedge B$	$(A \wedge B) \rightarrow X$	$X \rightarrow T$	$(A \wedge B) \rightarrow T$
T	T	T	T	T	T	T	T
T	T	T	F	T	T	F	F
T	T	F	T	T	F	T	T
T	T	F	F	T	F	T	F
T	F	T	T	F	T	T	T
T	F	T	F	F	T	F	T
T	F	F	T	F	T	T	T
T	F	F	F	F	T	T	T
F	T	T	T	F	T	T	T
F	T	T	F	F	T	F	T
F	T	F	T	F	T	T	T
F	T	F	F	F	T	T	T
F	F	T	T	F	T	T	T
F	F	T	F	F	T	F	T
F	F	F	T	F	T	T	T
F	F	F	F	F	T	T	T

Table I-1 — Unit I truth table.

8. Interpretation of Results

Validity means no false alarm when premises are adopted; 9 vacuous rows are desired no-alarm behaviour, the single fired row is the positive pathway animated in the Symptom Checker.

9. Practical Application

Deployed as the Symptom Checker card: toggles for A/B drive the chain live with inference strip $(A \wedge B) \rightarrow X \rightarrow T$.

10. Innovation / Proposed Improvement

- Add temporal logic and probabilistic thresholds.
- Surface the trace for clinician override.

11. Conclusion

The triage rule is sound, complete for its fragment, and explainable.

12. References

- Rosen, Discrete Mathematics, 8th ed.
- EU AI Act & FDA SaMD guidance.

PART II

Unit II — Combinatorics

Twelve-section standalone report for Unit II, covering counting of symptom sets.

1. Title of the Case Study (Unit II)

Unit II — Combinatorics: Sizing the Search Space for a 12-Symptom Triage Engine (Exactly, At Least, Ordered).

2. Problem Statement (Unit II)

For $n=12$, $k=5$: how many distinct sets under exactly five, at least five, and ordered five? Under- vs over-counting drives storage and UI.

3. Real-world Context

A 12-item checklist narrowed to five for rapid screening — planners must know test-case counts and whether ranking UI is worth 120× cost.

4. Mathematical Concept Used

$C(n,k)$, $P(n,k)$, $2^n = \sum C(n,k)$, complement principle, $P(n,k) = C(n,k) \cdot k!$.

5. Mathematical Formulation

Exactly-k: $C(12,5)$. At-least-k: $\sum_{i=5}^{12} C(12,i) = 2^{12} - \sum_{i=0}^4 C(12,i)$. Ordered-k: $P(12,5)$.

6. Analysis / Solution

$C(12,5) = 792$. At least five: $4096 - 794 = 3302$ ($792 + 924 + 792 + 495 + 220 + 66 + 12 + 1$). Ordered: $P(12,5) = 95,040 = 792 \cdot 120$.

7. Diagram / Tables

Policy	Formula	Count	Interpretation
Exactly 5	$C(12,5)$	792	One 5-set
At least 5	$\sum C(12,k)$	3,302	Any superset of 5
Ordered 5	$P(12,5)$	95,040	5 ranked

8. Interpretation of Results

3,302 bounds curation; 792 is UI choice count for fixed-5; 95,040 warns ranking needs virtualized rendering.

9. Practical Application

Combination Calculator panel with n/k sliders defaulting to 12/5, live exact integer arithmetic.

10. Innovation / Proposed Improvement

- Scale to $n \approx 50$ with inclusion–exclusion.
- Weighted counting with frequency priors.

11. Conclusion

Exact counting keeps the system auditable and the UI honest.

12. References

- Rosen, Chap. 6 — Counting.

Unit III — Graphs & Isomorphism

Twelve-section standalone report for Unit III, covering bipartite graphs and isomorphism certificates.

1. Title of the Case Study (Unit III)

Unit III — Graphs & Isomorphism: Bipartite Knowledge Graphs and Permutation-Similarity Certificates.

2. Problem Statement (Unit III)

Model knowledge as bipartite $G=(S \cup D, E)$ and prove two drawings G, G' with different layouts are isomorphic.

3. Real-world Context

Editors maintain symptom–disease edges in different tools; merging requires isomorphism proof, not layout trust. Enables 2-hop diagnosis: patient $\rightarrow$ symptom $\rightarrow$ disease.

4. Mathematical Concept Used

Bipartite graphs, degree sequences, biadjacency matrices, permutation similarity $A_2 = P \cdot A_1 \cdot P^T$.

5. Mathematical Formulation

$V = S \cup D$, $S = \{S_1 \text{ Fever}, S_2 \text{ Cough}, S_3 \text{ Rash}, S_4 \text{ Fatigue}\}$, $D = \{D_1 \text{ Influenza}, D_2 \text{ Measles}, D_3 \text{ Anemia}\}$,
 $E = \{S_1 D_1, S_1 D_2, S_2 D_1, S_3 D_2, S_4 D_1, S_4 D_3\}$, $|V| = 7, |E| = 6$. G' reorders as $(S_4, S_3, S_2, S_1 \mid D_3, D_2, D_1)$.

6. Analysis / Solution

Degrees: $S_1=2, S_2=1, S_3=1, S_4=2, D_1=3, D_2=2, D_3=1$; sequence $(3, 2, 2, 2, 1, 1, 1)$, sum $12 = 2 \cdot |E|$. Bijection $f(S_i) = S'_i, f(D_j) = D'_j$ preserves all six edges; reversal perms $r_S = [3, 2, 1, 0], r_D = [2, 1, 0]$ give $A_2 = P \cdot A_1 \cdot P^T$.

7. Diagram / Graph

Edge list:

#	Edge	Clinical meaning
1	$S_1 - D_1$	Fever indicates Influenza
2	$S_1 - D_2$	Fever indicates Measles
3	$S_2 - D_1$	Cough indicates Influenza
4	$S_3 - D_2$	Rash indicates Measles
5	$S_4 - D_1$	Fatigue indicates Influenza
6	$S_4 - D_3$	Fatigue indicates Anemia

Invariant	Rep.1	Rep.2	Match
Vertices $ V $	7	7	✓
Edges $ E $	6	6	✓
Degree sequence	(3,2,2,2,1,1,1)	(3,2,2,2,1,1,1)	✓
Bipartite	yes	yes	✓

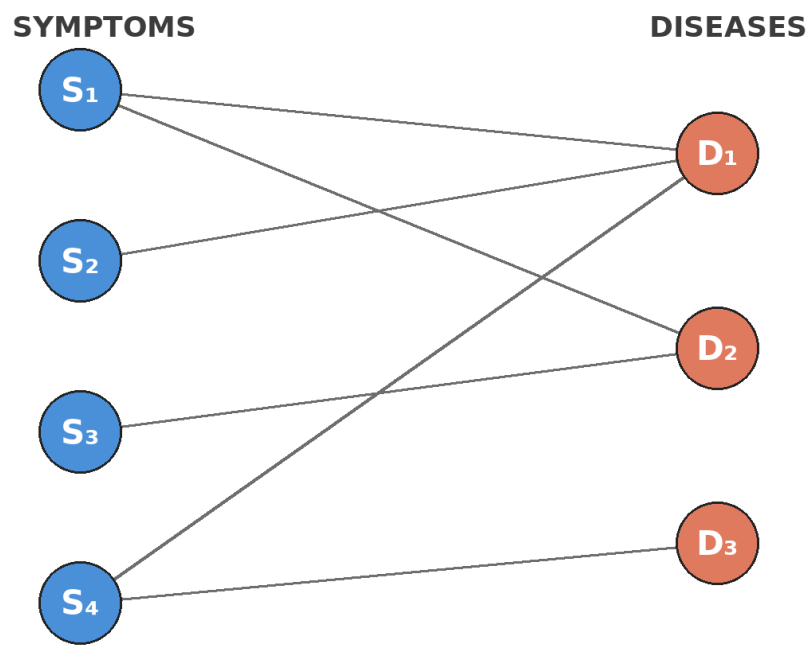


Figure III-1 — Bipartite layout.

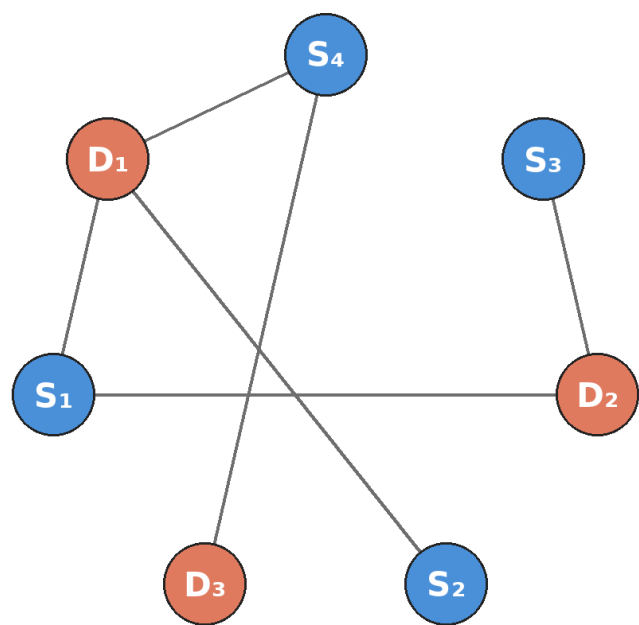


Figure III-2 — Circular scrambled layout.

Biadjacency matrices:

	D₁	D₂	D₃
S ₁	1	1	0
S ₂	1	0	0
S ₃	0	1	0
S ₄	1	0	1
	D₃	D₂	D₁
S ₄	1	0	1
S ₃	0	1	0
S ₂	0	0	1
S ₁	0	1	1

8. Interpretation of Results

Identical invariants plus exhibited P-matrix prove one knowledge base; layout irrelevant.

9. Practical Application

Isomorphism Explorer: toggle bipartite  circular with spring, click vertex to highlight incident edges and degree.

10. Innovation / Proposed Improvement

- Graph DB with adjacency lists for concurrent-edit merging.
- 3-partite patient-symptom-disease network.

11. Conclusion

Bipartite modelling plus permutation certificate turns visual intuition into a checkable artefact.

12. References

- Rosen, Chap. 10 — Graphs.